

Section 1: FD/Spectral operators — OPERATOR_REGISTRY in operators.py, available to <i>all</i> solver backends				
Operator	Math	Fourier multiplier	Modal	FD/Spectral
identity	1	1	✓	✓
laplacian	∇^2	$-\sum_i k_i^2$	✓	✓
laplacian_x	∂_x^2	$-k_x^2$	✓	✓
laplacian_y	∂_y^2	$-k_y^2$	✓	✓
laplacian_z	∂_z^2	$-k_z^2$	✓	✓
gradient_x	∂_x	ik_x	✓	✓
gradient_y	∂_y	ik_y	✓	✓
gradient_z	∂_z	ik_z	✓	✓
cross_deriv_xy	$\partial_x \partial_y$	$-k_x k_y$	✓	✓
cross_deriv_xz	$\partial_x \partial_z$	$-k_x k_z$	✓	✓
cross_deriv_yz	$\partial_y \partial_z$	$-k_y k_z$	✓	✓
biharmonic	∇^4	$(\sum_i k_i^2)^2$	✓	✓
derivative_3_x	∂_x^3	$-ik_x^3$	✓	✓
derivative_3_y	∂_y^3	$-ik_y^3$	✓	✓
derivative_3_z	∂_z^3	$-ik_z^3$	✓	✓

Section 2: Modal time-derivative classification — _OPERATOR_DECOMP in modal.py, constant-coefficient modal path only			
Operator	Math	Role in per-mode matrix	Time order
<i>Velocity (time order = 1) — damping / velocity coupling</i>			
first_deriv_t	∂_t	velocity-row coupling (D matrix)	1
mixed_T1_S1x	$\partial_t \partial_x$	velocity coupling $\times ik_x$	1
mixed_T1_S1y	$\partial_t \partial_y$	velocity coupling $\times ik_y$	1
mixed_T1_S1z	$\partial_t \partial_z$	velocity coupling $\times ik_z$	1
<i>Acceleration (time order = 2) — kinetic / mass matrix</i>			
d2_t	∂_t^2	off-diagonal mass matrix M	2
mixed_T2_S1x	$\partial_t^2 \partial_x$	mass coupling $\times ik_x$	2
mixed_T2_S1y	$\partial_t^2 \partial_y$	mass coupling $\times ik_y$	2
mixed_T2_S1z	$\partial_t^2 \partial_z$	mass coupling $\times ik_z$	2
mixed_T2_S2x	$\partial_t^2 \partial_x^2$	mass coupling $\times -k_x^2$	2
mixed_T2_S2y	$\partial_t^2 \partial_y^2$	mass coupling $\times -k_y^2$	2
mixed_T2_S2z	$\partial_t^2 \partial_z^2$	mass coupling $\times -k_z^2$	2
<i>Jerk (time order = 3) — eliminated via EOM substitution</i>			
d3_t	∂_t^3	EOM reduction source	3
mixed_T3_S1x	$\partial_t^3 \partial_x$	EOM reduction $\times ik_x$	3
mixed_T3_S1y	$\partial_t^3 \partial_y$	EOM reduction $\times ik_y$	3
mixed_T3_S1z	$\partial_t^3 \partial_z$	EOM reduction $\times ik_z$	3
<i>4th-order time (v6 perturbative theories only — never in base pipeline)</i>			
d4_t	∂_t^4	perturbative correction source	4
mixed_T4_S1x	$\partial_t^4 \partial_x$	correction $\times ik_x$	4
mixed_T4_S1y	$\partial_t^4 \partial_y$	correction $\times ik_y$	4
mixed_T4_S1z	$\partial_t^4 \partial_z$	correction $\times ik_z$	4
mixed_T4_S2x	$\partial_t^4 \partial_x^2$	correction $\times -k_x^2$	4
mixed_T4_S2y	$\partial_t^4 \partial_y^2$	correction $\times -k_y^2$	4
mixed_T4_S2z	$\partial_t^4 \partial_z^2$	correction $\times -k_z^2$	4

Section 1: 15 FD/spectral operators available to all backends (IDA, CVODE, Leapfrog, scipy, modal).
 Section 2: 22 temporal classification patterns (base) + 7 v6-only = 37 total in _OPERATOR_DECOMP.
 Temporal operators are not FD-stencil quantities — they enter the per-mode evolution matrix as kinetic (M), damping (D), or source terms handled by the Padé expm path (path D).
 FD stencil orders 2/4/6 available. Spectral mode uses exact multipliers for all-periodic BCs.